

Expectation Maximization Algorithm.

Recall,

$$F_{\theta}(q) = \mathbb{E}_{q(z)} \log \frac{p_{\theta}(x, z)}{q(z)}$$

The optimal $q(z) = p_{\theta}(z|x)$

suppose we evaluate $F_{\theta}(q)$ at some θ^t

$q(z)$ & $F_{\theta}(q)$ can be evaluated at different values of parameter, θ .

EM algorithm:

Initialize θ^1

For $t=1$ to convergence do:

$$q^{t+1}(z) = p_{\theta^t}(z|x)$$

compute $F_{\theta}(q^{t+1}) = \mathbb{E}_{q^{t+1}} \log \frac{p_{\theta}(x, z)}{q^{t+1}}$

E-step

M-step :

$$\text{Find } \theta^{t+1} = \underset{\theta}{\operatorname{argmax}} F_{\theta}(q^{t+1})$$

End For.

NPTEL

NOTEL

Convergence of the EM algorithm.

Claim: With EM, $\ell(\theta^{t+1}) \geq \ell(\theta^t)$

proof: Consider θ^t

$$q^{t+1}(z) = p_{\theta^t}(z|x)$$

$$\ell(\theta^t) = F_{\theta}(q^{t+1})$$

In M step, $\theta^{t+1} = \operatorname{argmax}_{\theta} F_{\theta}(q^{t+1})$

$$\therefore F_{\theta}(q^{t+1})|_{\theta=\theta^{t+1}} > F_{\theta}(q^{t+1})|_{\theta=\theta^t}$$

$$\text{also } F_{\theta}(q^{t+1})|_{\theta=\theta^{t+1}} \leq \ell(\theta^{t+1}),$$

by Jensen's inequality.

$$\Rightarrow \ell(\theta^{t+1}) \geq \ell(\theta^t) \quad \square$$

EM algorithm for GMMs.

$$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$$

$$= \sum_j \alpha_j \cdot N(x; \mu_j, \Sigma_j)$$

$$= \sum_z p_{\theta}(z) \cdot p_{\theta}(x|z)$$

$$z \sim \text{Discrete} \{ \alpha_1, \alpha_2, \dots, \alpha_m \}, \quad p_{\theta}(z) = \alpha_j$$

$$p_{\theta}(x|z=j) \sim N(x; \mu_j, \Sigma_j)$$

$$\theta = \left\{ \alpha_1, \alpha_2, \dots, \alpha_m, \mu_1, \mu_2, \dots, \mu_m, \right. \\ \left. \Sigma_1, \Sigma_2, \dots, \Sigma_m \right\}$$

$0 \leq \alpha \leq 1, \quad \mu \in \mathbb{R}^d, \quad \Sigma \in \mathbb{R}^{d \times d}$

EM for GMM:

initialize θ^t

Compute

$$q^{t+1}(z) = p_{\theta^t}(z|x)$$

$$= \frac{p_{\theta^t}(x|z) \cdot p_{\theta^t}(z)}{\sum_j p_{\theta^t}(x|z) p_{\theta^t}(z)}$$

$$= \frac{N\left(x; \mu_j^t, \Sigma_j^t\right) \cdot \alpha_j^t}{\sum_j \left(\text{denom} \right)}$$

= denom

Compute $F_{\theta}(q^{t+1})$

$$= \mathbb{E}_{q^{t+1}} \log \frac{p_{\theta}(x, z)}{q^{t+1}}$$

$$= \mathbb{E}_{p_{\theta^t}(z|x)} \log \frac{p_{\theta}(x, z)}{p_{\theta^t}(z|x)}$$

$$= \sum_{j=1}^M \left[\log \frac{p_0(x|z) \cdot p_0(z)}{p_{\theta^t}(z|x)} \right] p_{\theta^t}(z=j|x)$$

$$F_{\theta}(q^{t+1}) = \sum_{j=1}^M \log \left[\frac{N(x; \mu_j, \Sigma_j) \cdot \alpha_j}{(\text{denom})} \right] p_{\theta^t}(z=j|x)$$

$$\mu_j^{t+1} = \operatorname{argmax}_{\mu_j} F_{\theta}(q^{t+1})$$